PRN: 25070521161

NAME : Nakul Jibhakate

<u>EX: 4.4.1 SET OPERATIONS</u>

#ALGORITHM:

Step 1: Start

Step 2: Input Set A

Step 3: Convert the input values into Set A

Step 4: Input Set B

Step 5: Convert the input values into Set B

Step 6: Find the Union of Set A and Set B

Union = A | B

Step 7: Find the Intersection of Set A and Set B

Intersection = A & B

Step 8: Find the Difference of Set A and Set B

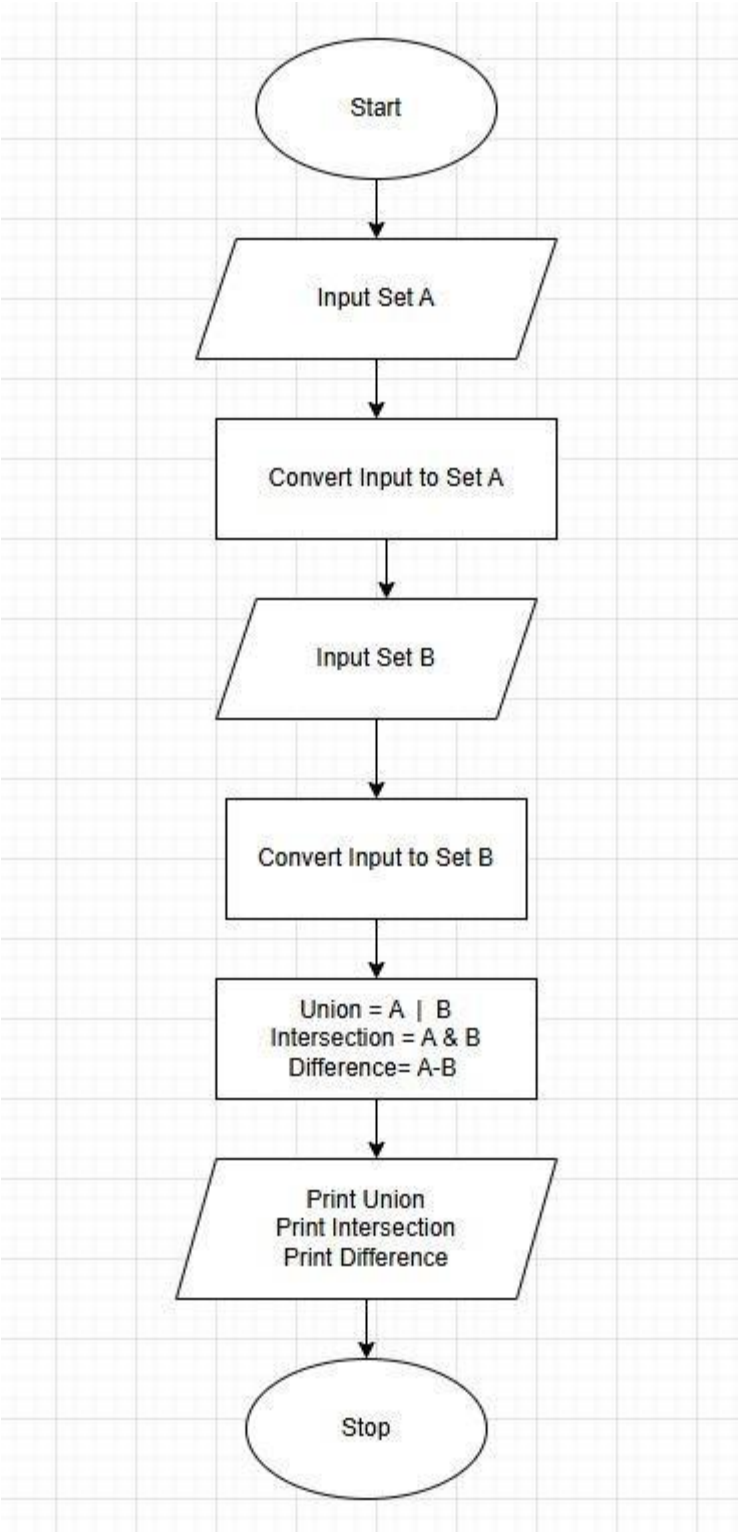
Difference = A − B

Step 9: Print the Union, Intersection, and Difference

Step 10: Stop

#CODE:

```
set_a = set(map(int, input("Set A: ").split())) set_b
= set(map(int, input("Set B: ").split()))
print("Union:", set_a | set_b)
print("Intersection:", set_a & set_b)
print("Difference:", set_a - set_b)
```


Start
Input Set A
Convert Input to Set A
Input Set B
Convert Input to Set B
Union = A | B
Intersection = A & B
Difference= A-B
Print Union
Print Intersection
Print Difference
Stop

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nakul.jibhakate.batch2025@sitnagpur.siu.edu.in Support Logout

4.1.1. Set Operations

00:24

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat...

Submit

Debugger

```python
# Type Content here...\\
setA = set(map(int, input("Set A: ").split()))
setB = set(map(int, input("Set B: ").split()))

print("Union:", setA.union(setB))
print("Intersection:", setA.intersection(setB))
print("Difference:", setA.difference(setB))
```

Average time	Maximum time
0.007 s	0.014 s
7.00 ms	14.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 6 ms

Debug

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test cases

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